pfi

Python for Fixed Income

COURSE 5C · RISK MODELS

Outliers: Robust Rules

WEEK 2 · DAY 1

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Today: a print to check

The synthetic blotter, September 2026

374

invented trades

14

planted in a copy

The untouched file is never changed; comparing the copy with it gives the labels.

Today's question

Which rule finds the bad prints without being told where they are?

And how many good prints does it send for checking along the way?

The labels only score. Every threshold comes from a published rule, chosen before any label is read.

Names for a flag

A price far off is a print to check: a typo, a wrong bond, a stale reference, or a real trade on a strange day.

Allowed names

a flag: a print to check

an unusual day, as history

a score, against planted labels

Never called

mispriced, rich, or cheap

an opportunity or a signal

a correction by itself

The rule's job is to put the print in front of a person.

How planting works

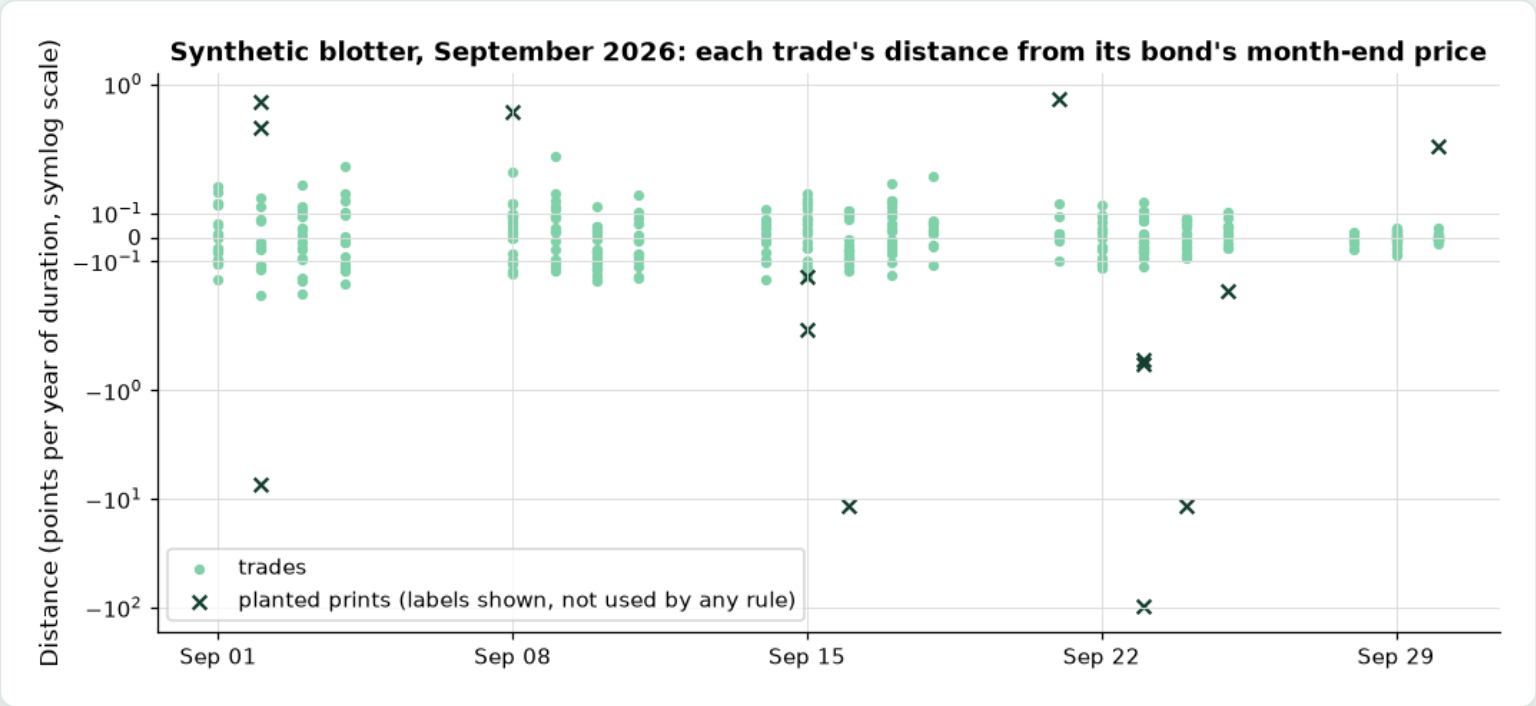
The problem	A screen can only be scored where the bad prints are known
A copy	Copy the blotter; the file itself is never changed
Off-market	Ten trades moved 1.5 to 4 points up or down, by a seeded generator
Decimal slipped	Four prices divided by ten: 102.4 typed as 10.24
The labels	The copy compared with the untouched file: used only to score

The planted prints

Trade	Date	Price in copy	True price	Kind
T0032	2026-09-02	10.244	102.435	decimal slipped
T0033	2026-09-02	91.908	88.513	off-market
T0035	2026-09-02	93.411	90.744	off-market
T0075	2026-09-08	111.296	109.410	off-market
T0159	2026-09-15	92.414	94.024	off-market
T0160	2026-09-15	102.775	106.702	off-market
T0196	2026-09-16	10.431	104.313	decimal slipped
T0238	2026-09-21	105.388	103.401	off-market
T0257	2026-09-23	98.174	101.907	off-market
T0275	2026-09-23	83.746	87.192	off-market
T0281	2026-09-23	10.134	101.337	decimal slipped
T0285	2026-09-24	9.168	91.675	decimal slipped
T0315	2026-09-25	89.533	91.919	off-market
T0363	2026-09-30	107.883	104.804	off-market

Every planted print is shown. The rules are scored on whether they find these rows without being told.

A distance from a reference



The blotter records trades, not mids. The reference is the bond's month-end price in the book; the distance is in points per year of duration.

The crosses are drawn so you can see them. No rule uses them.

How the z-score masks

The z-score	(x - mean) / standard deviation; |z| above 3 is a common flag
Built in	The mean and standard deviation are computed from all values, outliers included
Swelling	A few wild values pull the mean and swell the standard deviation
Masking	Every z-score shrinks, the outliers' too: they hide themselves and each other
What comes out	A score that cannot be trusted to find the values it hunts

Masking in numbers

5.168

standard deviation, planted prints in

0.090

without them (never known in practice)

1

trade with |z| above 3

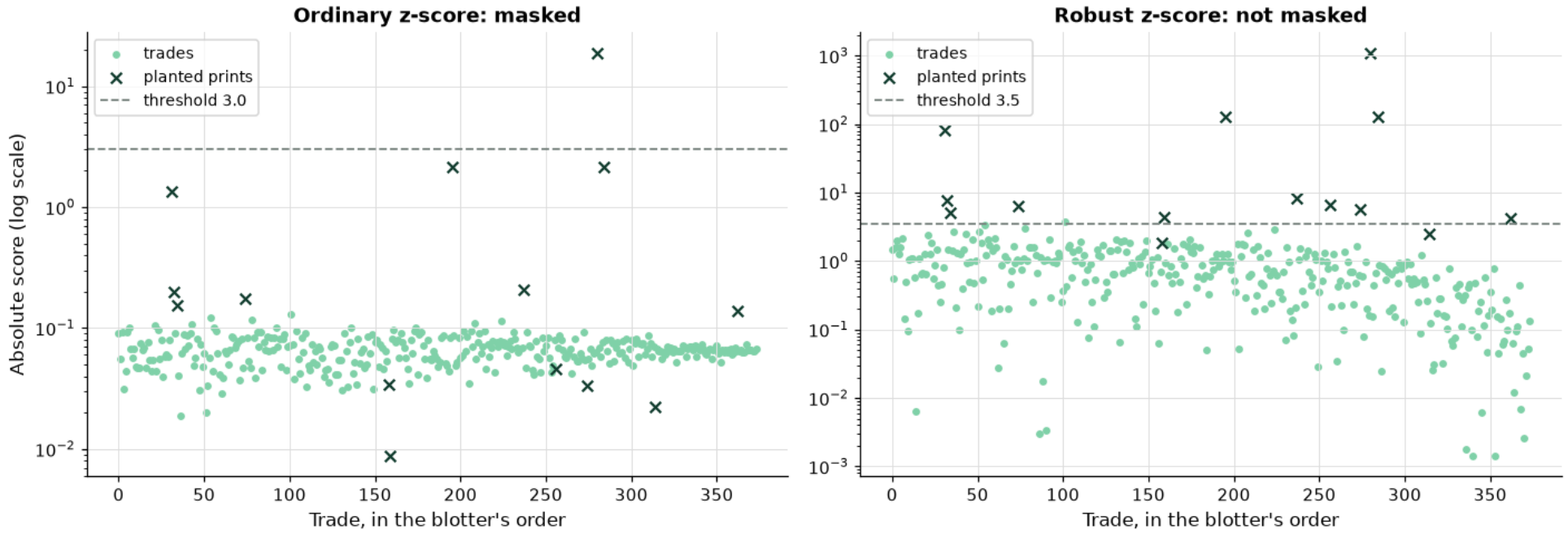
The four slipped decimals swell the yardstick 57 times. The one trade above 3 is the most extreme slip: it hides the other three, and every off-market print.

The outliers hide each other.

How the robust z-score works

Median	In place of the mean: up to half the values can be wild before it moves far
MAD	The median of each value's distance from the median, in place of the standard deviation
1.4826	Scales the MAD to equal the standard deviation for normal data
The rule	Flag |robust z| above 3.5: the NIST handbook, after Iglewicz and Hoaglin
What comes out	A score per value that its outliers cannot mask (robust_z)

Robust against ordinary



0.0887

scaled MAD: the robust yardstick

13

trades flagged at 3.5

Tukey's fences

```
flag x  <  Q1 - k x IQR   or   x  >  Q3 + k x IQR          IQR = Q3 - Q1
```

17

outside the inner fences, k = 1.5

10

outside the outer fences, k = 3

For normal data the inner fences sit about 2.7 standard deviations from the median and the outer fences about 4.7: looser and stricter than 3.5.

Tukey, Exploratory Data Analysis, 1977: the box plot's rule for what to draw apart.

How precision and recall work

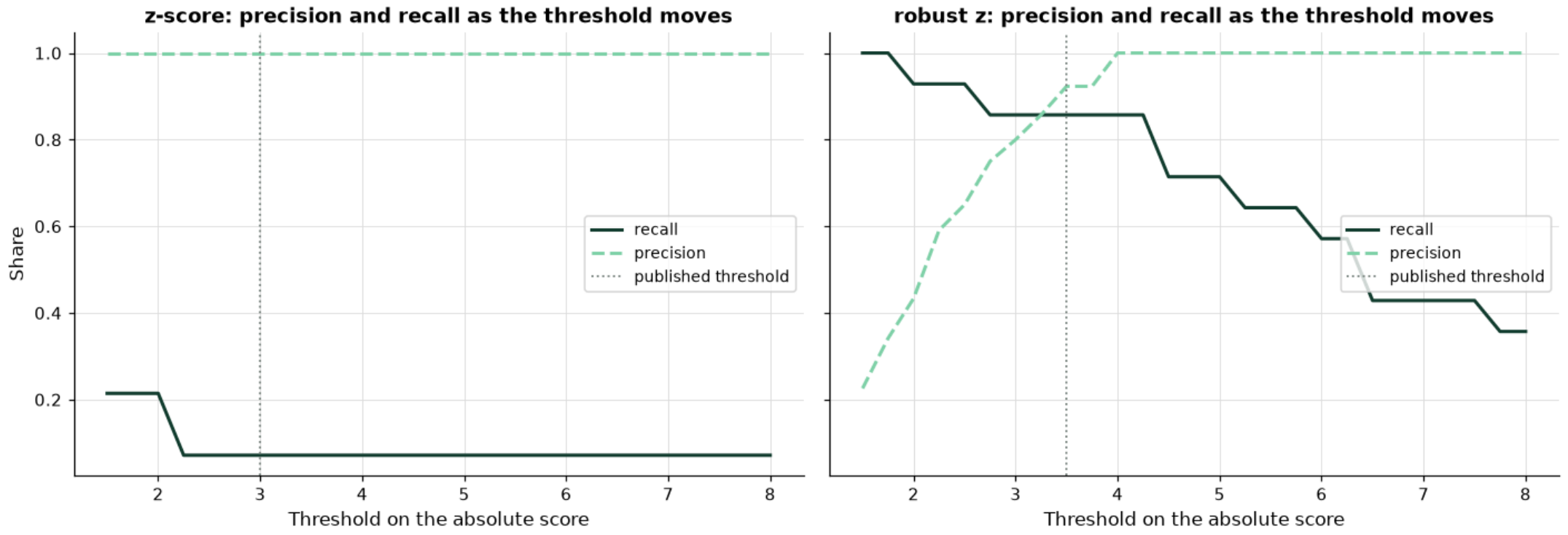
Two mistakes	A screen flags a good print, or misses a bad one
Precision	The share of flags that were bad prints: the checker's time well spent
Recall	The share of bad prints that were flagged: how few slip through
The trade-off	A higher threshold flags less: precision usually rises, recall falls
The labels	Here the planted ones; they score, and choose nothing (precision_recall_at)

Every rule, scored

Rule	Flags	Planted	Precision	Recall
z-score, |z| above 3	1	1	100%	7%
robust z, above 3.5 (NIST)	13	12	92%	86%
inner fences (k = 1.5)	17	12	71%	86%
outer fences (k = 3)	10	10	100%	71%

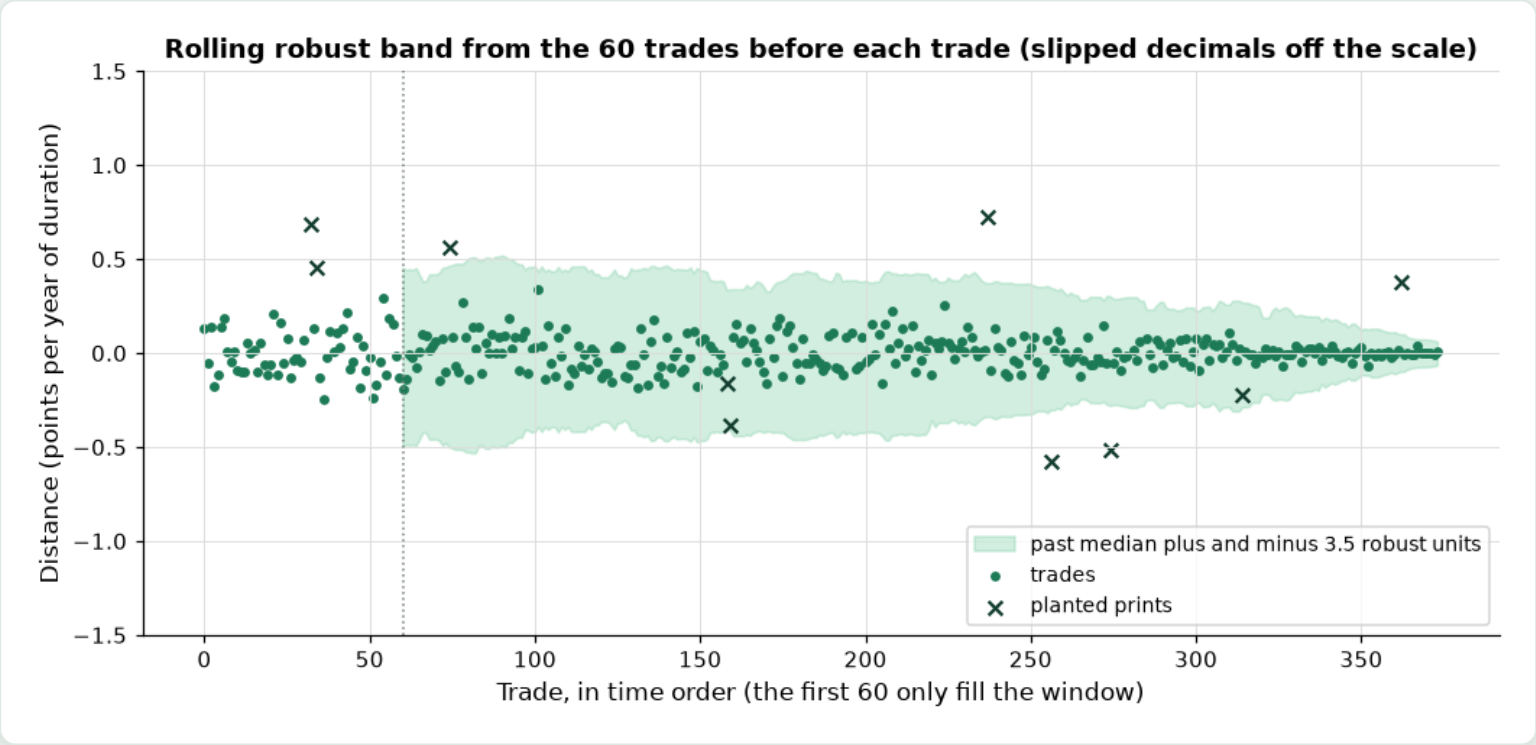
The masked z-score finds one slipped decimal. The robust rules find most of the prints.

The threshold sweep



No threshold rescues a masked score. And no threshold here is chosen by reading this chart.

Rolling, past trades only



8

flags, of 314 scored

73%

recall, 11 planted

Precision 100%, from what each trade had when it was booked. The first 60 trades only fill the window.

The 3.5 rule and its warning

NIST/SEMATECH e-Handbook of Statistical Methods, section 1.3.5.17

"Detection of Outliers"

Modified Z-score: 0.6745 (x - median) / MAD, after Iglewicz and Hoaglin

"These authors recommend that modified Z-scores with an absolute value of greater than 3.5 be labeled as potential outliers." And: "masking is one reason that trying to apply a single outlier test sequentially can fail."

Source: itl.nist.gov/div898/handbook, section 1.3.5.17, citing Iglewicz and Hoaglin, How to Detect and Handle Outliers, 1993.

Excel side by side

Excel side by side (tested in Excel)

`=MEDIAN(A2:A375)`	median -0.0012
`=MEDIAN(ABS(A2:A375-MEDIAN(A2:A375)))`	MAD 0.0598
`=(A2-$E$1)/(1.4826*$E$2)`	robust z 1.499
`=QUARTILE.INC(A2:A375,1)`	Q1 -0.0606
`=SUMPRODUCT(--(ABS(C2:C375)>3.5))`	13 flags at 3.5
`=SUMPRODUCT(--(ABS(C2:C375)>3.5),B2:B375)/SUM(B2:B375)`	recall 86%

The MAD is one array formula: MEDIAN of ABS of the range minus its MEDIAN. Every value matched Python.

Save the day: what changed since week 1

1. Start of today

Commit the module as day 5 left it

2. git diff --stat

Before adding: what changed in the folder

3. Commit

Today's three functions and their tests

```
git diff --stat week1-riskkit
```

In your own folder: compare with the tag you set at the end of week 1.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

What is drawn, from which inputs, in which unit: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: a robust z that forgets the 1.4826: standard deviation 1.496 on normal data, not 1.

Code that runs is not the same as code that is right.

What you can do now

- Plant bad prints in a copy; keep labels for scoring only
- Measure distance from a reference, scaled by duration
- Show masking; score with robust_z at 3.5 and iqr_flags
- Score rules with precision_recall_at; read a sweep
- Score each trade against the past only; 28 tests pass

Week 2, Day 2 Isolation forest and local outlier factor